

AKTU Lucknow

MID-TERM EXAMINATION ? 2025

Fluid Mechanics ? Mid-Term Examination, 2025

Subject Code: ME304 Semester: 4 Regulation: R20

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Fluid Statics)

1. Derive the expression related to Fluid Statics and explain each term. [12 Marks]
2. List the advantages and limitations of Fluid Statics. Explain any two in detail. [12 Marks]
3. How is Fluid Statics implemented? Give an algorithm / procedure and explain. [8 Marks]

SECTION B (Unit 2: Kinematics of Fluid Flow)

4. Compare and contrast the key concepts involved in Kinematics of Fluid Flow. [12 Marks]
5. List the advantages and limitations of Kinematics of Fluid Flow. Explain any two in detail. [12 Marks]
6. Define Kinematics of Fluid Flow. Explain its significance in Fluid Mechanics with a suitable example. [12 Marks]

SECTION C (Unit 3: Bernoulli's Equation and Applications)

7. List the advantages and limitations of Bernoulli's Equation and Applications. Explain any two in detail. [12 Marks]
8. Derive the expression related to Bernoulli's Equation and Applications and explain each term. [12 Marks]
9. Compare and contrast the key concepts involved in Bernoulli's Equation and Applications. [12 Marks]

SECTION D (Unit 4: Flow Through Pipes)

10. Explain Flow Through Pipes with a neat diagram/sketch and a suitable example. [8 Marks]
11. List the advantages and limitations of Flow Through Pipes. Explain any two in detail. [8 Marks]
12. Solve a problem related to Flow Through Pipes. Show all the steps clearly. [8 Marks]

SECTION E (Unit 5: Boundary Layer Theory)

13. Solve a problem related to Boundary Layer Theory. Show all the steps clearly. [12 Marks]

Marks]

14. Discuss the engineering applications of Boundary Layer Theory. [12 Marks]

15. Derive the expression related to Boundary Layer Theory and explain each term. [12 Marks]

*** End of Question Paper ***